

SQL Query Practice Dataset (Subquery Practice)

DATABASE TABLES

1. Departments

```
CREATE TABLE departments (  
  dept_id INT PRIMARY KEY,  
  dept_name VARCHAR(50),  
  location VARCHAR(50)  
);  
  
INSERT INTO departments VALUES  
(1,'HR','New York'),  
(2,'IT','London'),  
(3,'Finance','Chicago'),  
(4,'Sales','Boston');
```

2. Employees

```
CREATE TABLE employees (  
  emp_id INT PRIMARY KEY,  
  emp_name VARCHAR(50),  
  dept_id INT,  
  manager_id INT,  
  salary INT,  
  hire_date DATE  
);  
  
INSERT INTO employees VALUES  
(101,'Amit',1,NULL,50000,'2020-01-15'),  
(102,'Ravi',2,101,70000,'2019-05-20'),  
(103,'Neha',2,101,65000,'2021-02-10'),  
(104,'Priya',3,102,80000,'2018-07-01'),  
(105,'Arjun',3,102,90000,'2017-09-15'),  
(106,'Karan',4,103,60000,'2022-03-18'),  
(107,'Meera',4,103,62000,'2021-11-12'),  
(108,'Sneha',2,101,72000,'2019-12-01'),  
(109,'Rahul',1,NULL,55000,'2023-04-08'),  
(110,'Vikas',3,104,85000,'2020-08-22');
```

3. Projects

```
CREATE TABLE projects (  
  project_id INT PRIMARY KEY,  
  project_name VARCHAR(50),  
  emp_id INT,  
  budget INT  
);
```

INSERT INTO projects VALUES

(1,'ERP',102,500000),
(2,'CRM',103,300000),
(3,'AI Platform',104,900000),
(4,'Analytics',105,700000),
(5,'HR Portal',107,200000),
(6,'Sales App',110,600000);

SQL QUERY PRACTICE QUESTIONS

Beginner Level

1. Find all employees earning more than the company's average salary.
2. Find the employee(s) with the highest salary.
3. Find employees who work in the same department as Priya.
4. Find employees who earn more than Rahul.
5. Find employees hired before Ravi.
6. Find all employees in the IT department.
7. Find the department name of the employee earning the highest salary.

Intermediate Level

8. Find employees earning more than their department's average salary.
9. Find employees who are managers.
10. Find employees who are not managers.
11. Find employees who have been assigned at least one project.
12. Find employees who have never been assigned to any project.
13. Find departments whose average salary is greater than the company's average salary.
14. Find employees earning more than every employee in the HR department.

Advanced Level

15. Find employees earning more than their manager.
16. Find the second-highest-paid employee in each department.
17. Find departments where every employee earns more than 60,000.
18. Find employees working in the department with the highest average salary.
19. Find employees whose salary is greater than the average salary of employees assigned to projects.
20. Find employees who do not belong to the department with the highest total salary expenditure.